

Introduction

In the digital age, the proliferation of information has intensified, consumers often find themselves overwhelmed by information, which often leads to the TL;DR paradigm. Language models offer capacity to control level of details for information. Yet, when condensing news media, it's imperative to hinge on narrative principles. Narratives, while varying across disciplines, are essentially interconnected events. Language models are good at decoding complex textual data, but the effectiveness of their narrative reduction needs to be critically assessed.

We argue for preserving human agency in this interpretive process through visual analyses, rather than passively accepting the narratives provided by algorithmic black boxes.

Abstract

In this work, we investigate the intersection of narrative theory and data visualisation in the realm of news media. We present an exploration of the narrative construction mechanisms and the efficacy of visualisation techniques in conveying news stories. A novel method is introduced to extract and visualize narratives from large corpora of news articles. We compare two visualisation techniques applied under different information levels to ascertain the nuanced ways in which narratives shape our comprehension and interpretation of news. Furthermore, we examine how visual aids can enhance or distort these narratives. The results of this research could offer fresh insights into how we decode and engage with the dense information landscape prevalent in modern news media.

Narrative maps

Recent research in the field of information cartography has introduced the concept of a “narrative map” [1], visualizing narratives as weighted directed acyclic graphs having a single starting event (source) and a single ending event (sink). The weights represent the probability of transitioning from one event to another, based on their coherence or logical consistency.

Our approach focuses on exploring the extent to which visuals can guide the reader through the narrative interpretations obtained by moving between the information pyramids. With this in mind, we have developed two distinct styles of narrative maps: the “Aggregated Map” and the “Atomic Map”. Both visually represent a collection of news articles to depict common narratives evolving over time, but at different levels of detail.

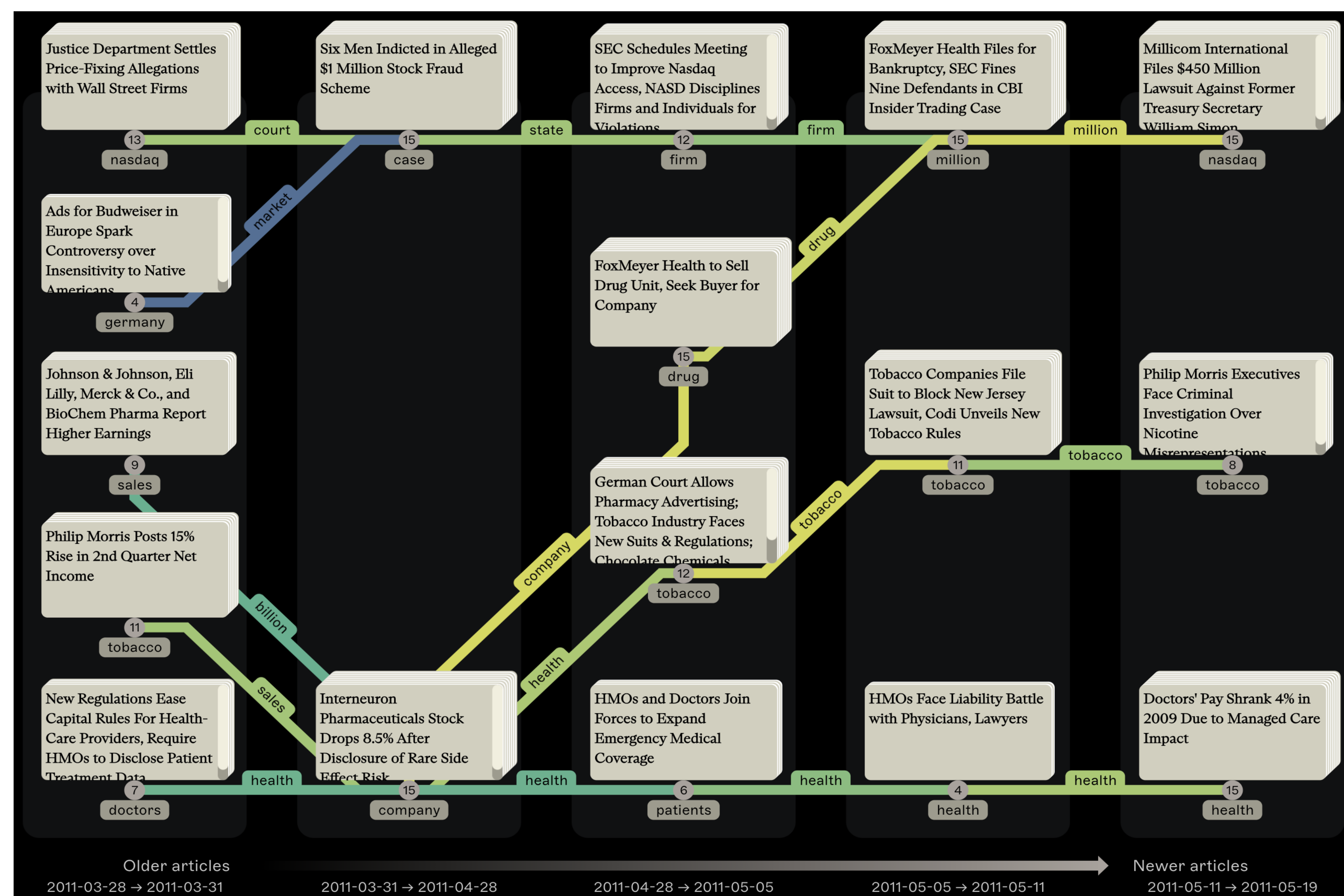


Figure 2. An example of our Narrative Maps, a novel method of visualizing the narrative structure of news articles (Aggregated Map).

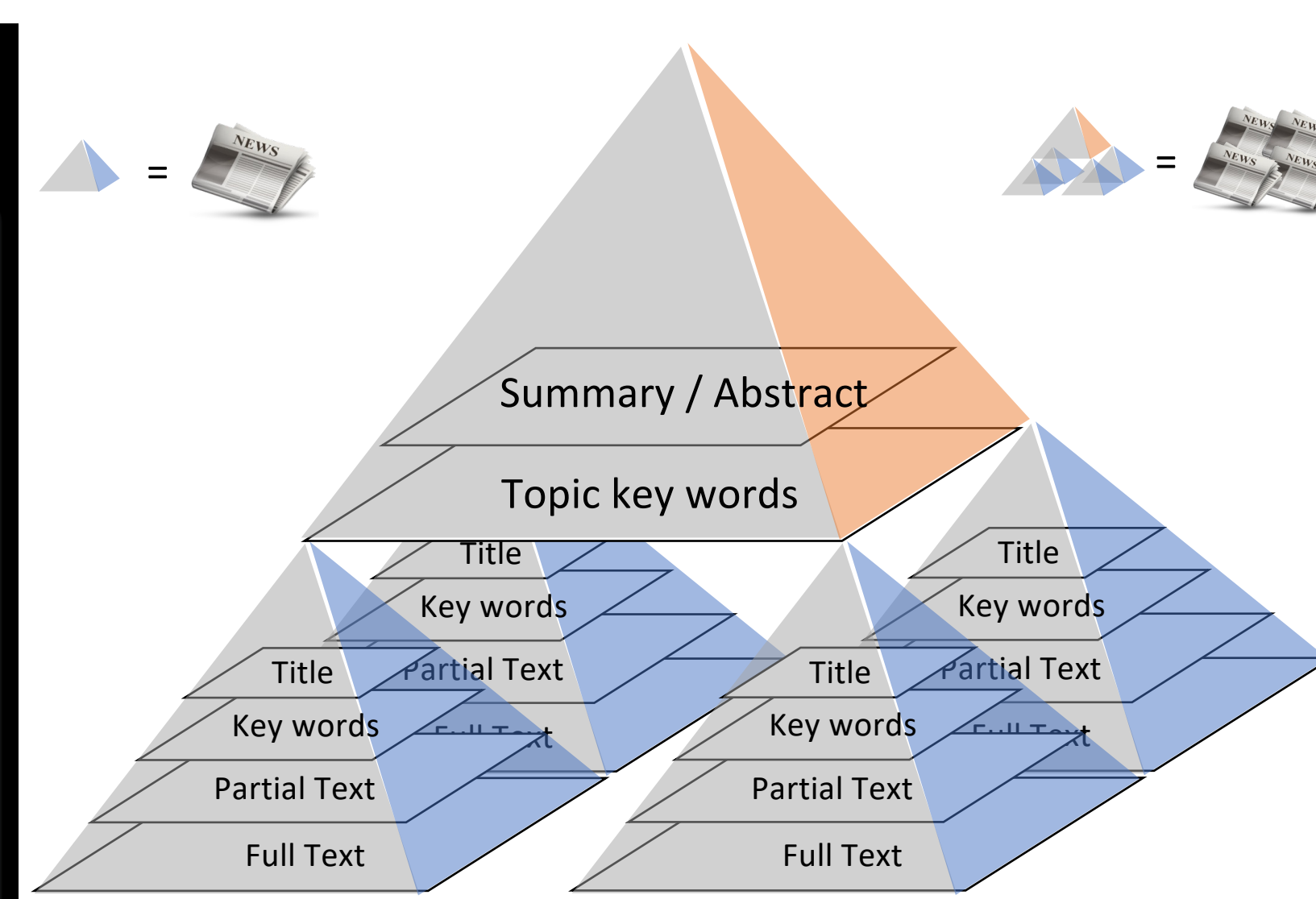


Figure 3. An example of our Narrative Maps (Atomic Map).

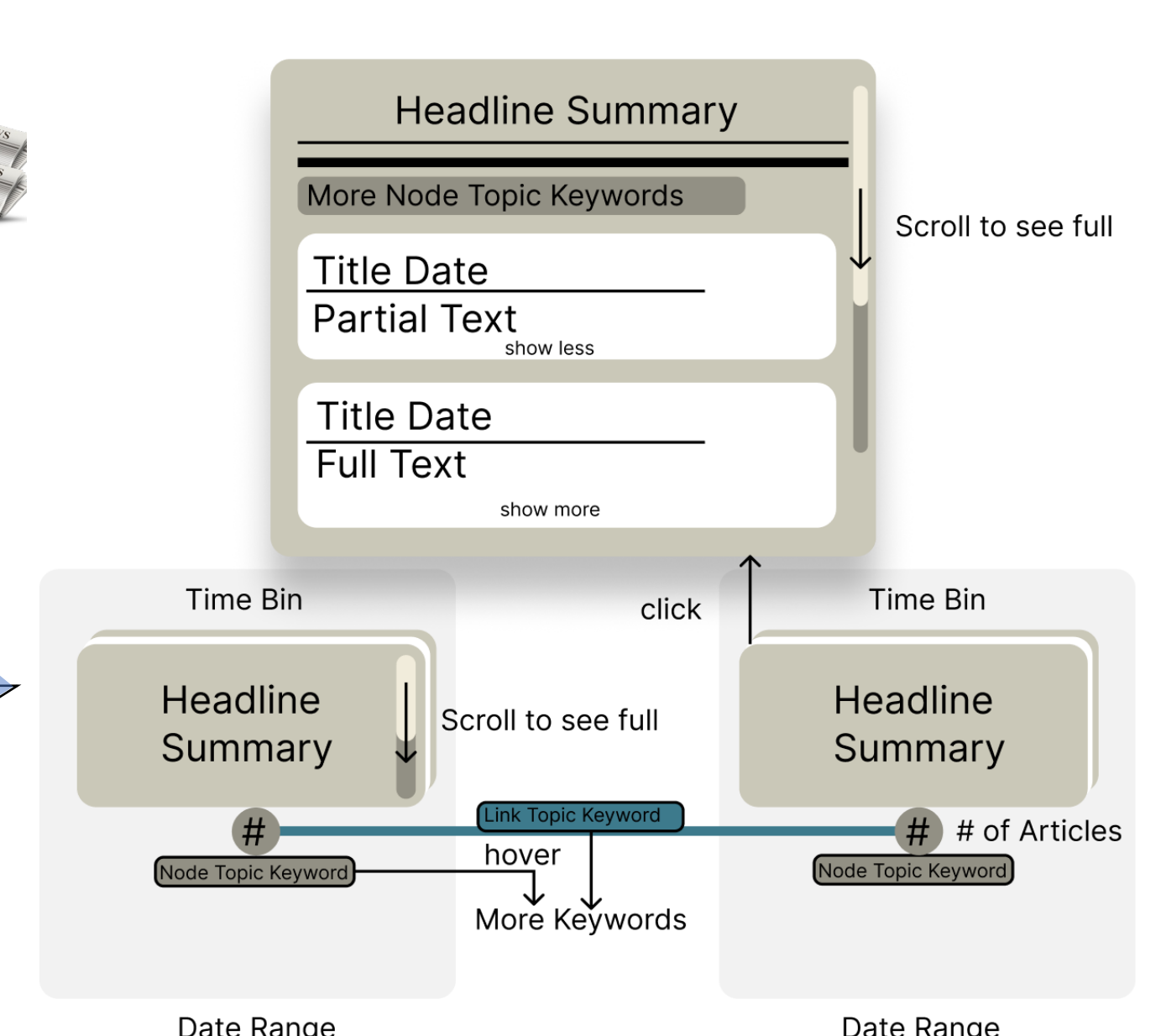


Figure 4. Aggregated Map design concept

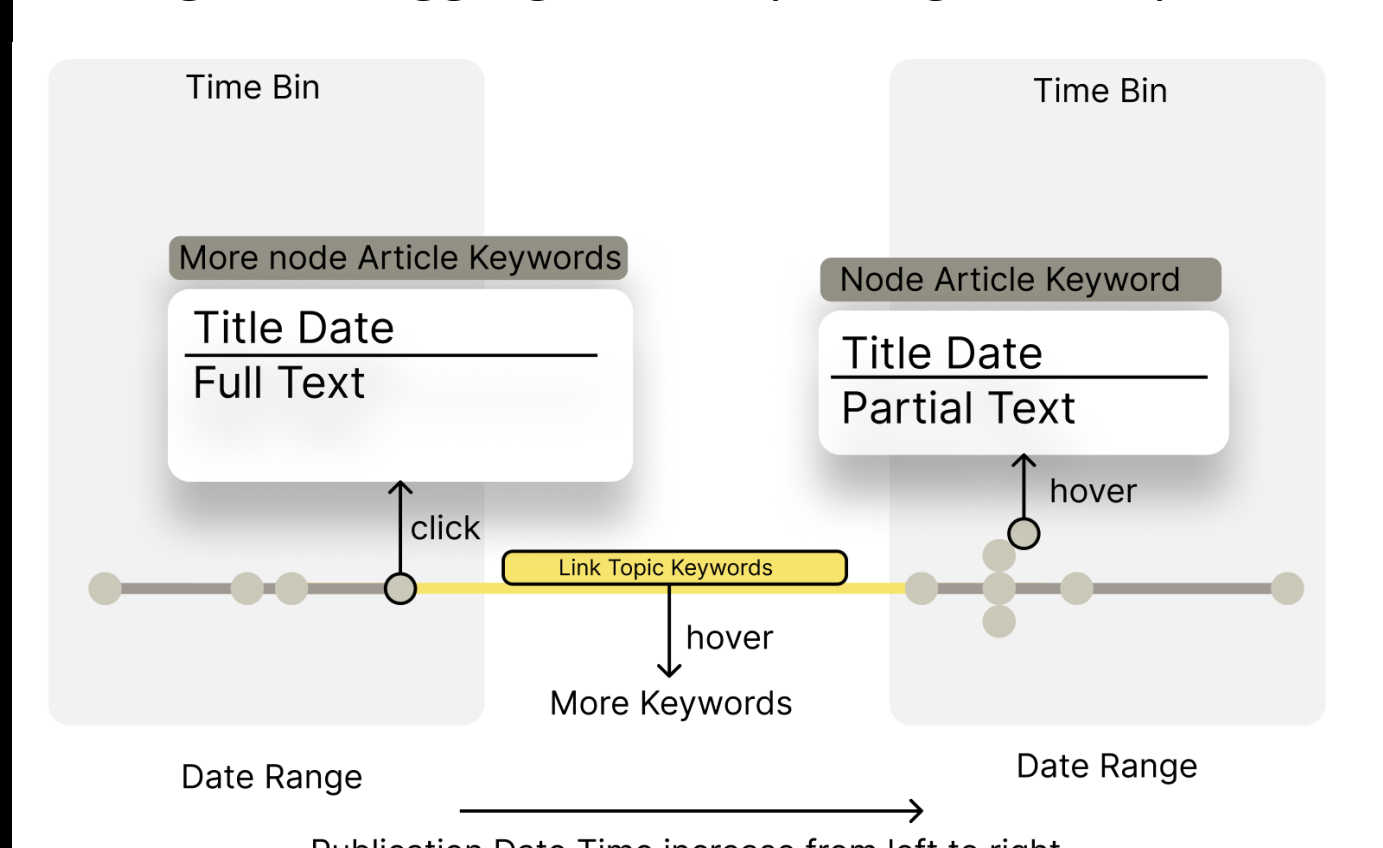


Figure 5. Atomic Map design concept

Narrative extraction

Our unique approach integrates entity-based and event-based methods to extract narrative structure from text collections.

1.Data Ingestion: We gather timestamped files from various sources, including fake and authentic news datasets.

2.Temporal Embedding Generation: We use the GPT-4 model to generate numerical text representations, integrating textual content and timestamps for a richer, time-semantic embedding.

3.Embedding Optimization and Semantic Grouping: We transform high-dimensional embeddings into lower-dimensional space using UMAP for better data visualization and analysis and group these embeddings using HDBSCAN.

4.Identification of Pivotal Temporal Inflection Points: We introduce a technique to form a structured narrative arc from temporal clusters, refine global clusters into local groupings, and select keywords to represent each cluster's themes.

5.Formation of Directed Acyclic Graph (DAG): We encapsulate the narrative trajectory in a DAG structure, leveraging Edmonds' Algorithm to find the most efficient pathways and ensure minimal linkage redundancy.

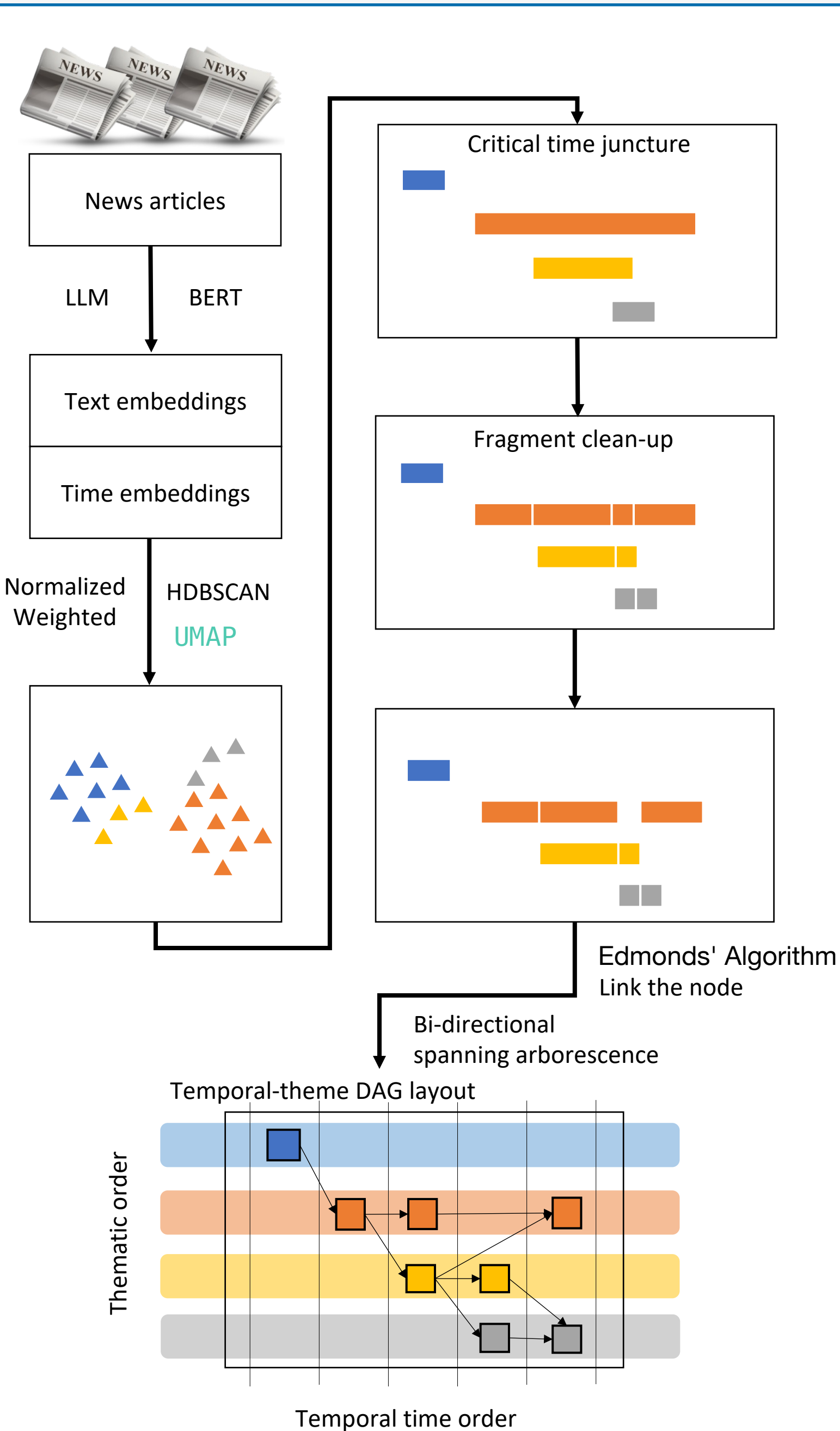


Figure 6. Narrative extraction method and visualisation layout.

Evaluation

Our study involved a two-phase user evaluation of narrative summaries derived from metro maps.

Expert Qualitative Evaluation

For the qualitative evaluation of the Narrative Map, expert participants were requested to generate text summaries from six metro maps using two variants: 'aggregated' and 'atomic'. The results showed a preference for the aggregated maps, indicating that users prefer information that is consolidated.

Leveraging LLM for Quantitative Evaluation

For the quantitative evaluation of Narrative Extraction Methods (NEM), we utilized Large Language Models (LLM) to create AI-generated narratives and associated visual graphs, which served as potential benchmarks. Our goal was to measure the effectiveness of various recovery methods, encompassing both a conventional algorithmic approach and a novel LLM-based method. This included comparing the resulting graphs and employing tools such as graph edit distance to determine similarity.

Conclusions & Future Directions

In future, we envision creating test datasets with set narrative structures, spanning a variety of topics, to rigorously and quantitatively assess and refine our narrative extraction techniques. Exploring LLM's Graph Creation Capacity from Real Datasets: Our next step is to delve deeper into the capabilities of LLM for graph generation from authentic datasets.

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